



Data Analytics Intern — Case Study

Project Brief | Cindy Chen | March 2026

1. Project Overview

Candidate	Cindy Chen
Role	Data Analytics Intern — KPay Australia
Submitted to	Luyin Qian (Louie), Growth and Partnerships Manager
Deadline	March 1, 2026
Dataset	200,000+ merchant records scraped from Google Maps
Tools	Python (pandas, phonenumbers, rapidfuzz), Jupyter Notebook, Tableau
Format	15-minute presentation covering all three parts

2. Business Problem

KPay launched in Australia in October 2025 targeting 2.65 million SMEs with a \$0 EFTPOS terminal, 1.4% transaction fee, same-day settlement, and multilingual support for Asian payment methods including UnionPay and JCB.

The BD team has been provided with a raw Google Maps scrape of 200,000+ merchant records. This dataset is not usable for outreach. It contains data quality issues, overseas records, uncontactable phone numbers, and no prioritization logic. Without cleaning and structuring, BD reps would waste significant time on unqualified or unreachable leads.

This case study addresses three questions:

- **Part 1** — How do we clean and enrich this dataset to make it BD-ready?
- **Part 2** — How do we priorities outreach, so BD reps focus on the highest-value merchants first?
- **Part 3** — How do we track the performance of this canvassing initiative over time?

3. Scope & Deliverables

Part	Description	Deliverable
Part 1	Data Cleaning & Enrichment — identify quality issues, clean the dataset, and propose enrichment data points to improve lead qualification	Cleaned CSV export + data quality audit documentation
Part 2	Sales Canvassing Strategy — segment merchants into tiers using a scoring model and design a concrete BD outreach playbook	Tiered merchant dataset + canvassing strategy framework
Part 3	Performance Dashboard — design a dashboard for the sales manager to track the canvassing initiative	Dashboard blueprint with 3–5 KPIs and rationale

4. Data Quality — Anticipated Issues & Planned Approach

Before analysis, the following data quality issues are expected in a raw Google Maps scrape of this size and will be addressed systematically:

4.1 Phone Number Quality

The phone column is the most critical field for BD outreach. The planned approach classifies each phone number across four dimensions:

- Frequency — any phone appearing more than 10 times across unique businesses is statistically a placeholder, not a real number
- Type classification — Australian numbers will be categorised into mobile, landline, shared_cost (1300), toll_free (1800), short_business (13xx), premium, and emergency types using the phonenumbers library
- Length validation — after stripping formatting, numbers not matching expected digit counts for their type are flagged as Level 2 suspicious rather than deleted
- State/area code consistency — NSW records with non-02 landline prefixes are soft-flagged (VoIP and number portability are legitimate exceptions)

Output: three new columns — phone_type, phone_quality (Level 1 / 2 / 3), and reachability_score (0–30 pts) — which feed directly into the KMF scoring model in Part 2.

4.2 Geographic Data Issues

Two geographic issues are anticipated:

- Non-Australian records — the state column contains US state codes (NY, CA, FL etc.), full Australian state names in inconsistent capitalisation, and garbage values. Resolution: a multi-signal classifier checks state value first, then interrogates the address column for AU/overseas signals, producing two output files: **Cleaned_AU_Data.csv** and **Out_of_AU_Data.csv**

- Missing suburb — approximately 50% of records are expected to have null suburb values. Resolution: regex-based extraction from the address column, cross-validated against an official 16,220-entry AU suburb reference list, targeting a fill rate above 70%

4.3 Sector Label Inconsistency

sector_level_1 is expected to contain 100+ unique labels representing variations of 5 core categories (F&B, Beauty & Wellness, Retail, Retail - Electronics, Professional Services). Additionally, a large 'Others' bucket and ~19% null values cannot be prioritised without reclassification.

Resolution: a two-pass remapping approach — keyword matching on sector_level_1 label variations, followed by keyword rescue of Others/null records using sector_level_3 content. Fuzzy matching (rapidfuzz) handles edge cases not covered by the seed keyword list.

4.4 Duplicate Records

Two duplicate checks will be applied: exact lead_key duplicates (hard deduplication) and business_name + suburb duplicates (flagged but not deleted, as franchise chains like The Lott legitimately appear multiple times across different locations).

5. Data Enrichment Proposals

Data enrichment in this project was performed directly on the existing dataset — extracting new signals from columns already present in the raw data, rather than sourcing external data. Six enrichment columns were engineered in Python and added to the cleaned dataset. Each column answers a specific BD question that the raw data could not answer on its own.

ENRICHMENT 1 — PHONE TYPE CLASSIFICATION

New column: phone_type

Source: Derived from the phone and lead_key columns using the Python phonenumbers library. Each phone number was parsed and classified into one of seven types: mobile, landline, shared_cost (13xx numbers), toll_free (1800), premium_rate, unknown, or short_business.

BD decision unlocked: Phone type drives the reachability score in the KMF model. A mobile number scores 30/30 (direct to decision maker). A landline scores 25/30. A shared_cost 13xx number scores 20/30 (likely a call centre, harder to reach the owner). A Level 1 fake phone scores 0 — the BD rep skips this lead entirely. Without this classification, every phone number would be treated as equally contactable.

ENRICHMENT 2 — PHONE QUALITY SCORING (3-LEVEL)

New column: phone_quality (Level 1 / Level 2 / Level 3)

Source: Derived from four signals computed within the dataset: (1) phonenumber library validity check, (2) cross-reference match score between the phone column and the phone embedded in lead_key, (3) frequency analysis flagging any phone appearing more than 10 times as a placeholder, and (4) bad pattern detection for numbers with 6+ trailing zeros or all-repeating digits.

BD decision unlocked: Level 1 phones (fake/placeholder) score 0 reachability — BD reps do not call these leads. Level 2 phones (suspicious but possibly real) score 3/30 — low priority. Level 3 phones (valid, unique, cross-matched) score up to 30/30 — these are the leads a BD rep dials first. This single column prevents BD reps from wasting time on approximately 0.2% completely fake leads and deprioritises the 56% suspicious numbers in the dataset.

ENRICHMENT 3 — REACHABILITY SCORE (0–30 PTS)

New column: reachability_score

Source: Computed from phone_quality and phone_type together. Level 1 = 0 pts regardless of type. Level 2 = 3 pts regardless of type. Level 3 mobile = 30 pts, Level 3 landline = 25 pts, Level 3 shared_cost = 20 pts, Level 3 toll_free = 10 pts. This single numeric score replaces the categorical quality and type columns in the KMF scoring formula.

BD decision unlocked: Feeds directly into the KMF model as the Reachability component (40 pts maximum when combined with the address flag). Converts qualitative phone assessment into a quantitative score that can be ranked and sorted. A BD rep's daily call list is sorted by reachability_score descending — the most contactable merchants always appear first.

ENRICHMENT 4 — DATA COMPLETENESS FLAGS

New columns: has_address (boolean), has_sector (boolean)

Source: Boolean flags derived directly from null-checking the address and sector_level_1 columns. has_address = True if address is not null. has_sector = True if sector_level_1 is not null.

BD decision unlocked: Both flags feed into the Data Completeness component of the KMF model (10 pts maximum). A merchant with a verified address gets 3 bonus points — they can be visited in person, not just called. A merchant with a confirmed sector gets 3 bonus points — their pitch angle is known. Records missing both fields score lower and are deprioritised for in-person BD visits.

ENRICHMENT 5 — KPAY RELEVANCE FLAG

New column: kpay_relevant (boolean)

Source: Derived from sector_level_3 using keyword exclusion matching. Businesses in categories that do not transact with consumers using EFTPOS — such as Retirement, Equestrian, NGO, Construction, Community Health, Religious, and Recycling — were flagged

as kpay_relevant = False and excluded from the scoring dataset. This flag was applied before KMF scoring, removing 11,194 records from the working dataset.

BD decision unlocked: Ensures the BD team never receives a lead that is structurally unsuitable for KPay — a construction contractor or religious organisation will never sign up for an EFTPOS terminal for consumer payments. Removing these records improves the signal quality of the entire dataset and prevents wasted BD effort.

ENRICHMENT 6 — NAME + SUBURB DUPLICATE FLAG

New column: name_suburb_dup (boolean)

Source: Derived from a compound duplicate check on business_name + suburb together. Records flagged True share an identical name and suburb as a previous record in the dataset. These were deliberately flagged rather than deleted — franchise chains such as The Lott legitimately appear multiple times across different locations, and hard deletion would remove valid leads.

BD decision unlocked: BD manager can filter name_suburb_dup = True to review suspected duplicates before assigning territory. This prevents two BD reps from calling the same business independently, which would damage KPay's professional reputation with the merchant.

6. Merchant Segmentation — KMF Scoring Model

Merchants will be scored and tiered using the KPay Merchant Fit (KMF) Index — a weighted composite score built specifically around KPay's revenue model and competitive positioning.

6.1 Model Design

Component	Weight	Rationale
Reachability	40 pts	An unreachable lead costs the same BD time as a real one but produces zero revenue. Phone type and quality determine whether a rep can actually reach the decision maker.
Sector Priority	35 pts	KPay earns on transaction volume. F&B and Beauty & Wellness businesses process dozens of daily payments vs. a consultant who invoices monthly. Sector determines revenue potential.
Geo Efficiency	15 pts	KPay's multilingual support and UnionPay/JCB acceptance is strongest in Asian-Australian community suburbs. Geo score also enables territory clustering for BD rep efficiency.
Data Completeness	10 pts	Records with verified addresses and sector data are more actionable. This component rewards data quality and penalises records where critical fields are missing.

6.2 Tier Assignment

KMF scores will be mapped to three tiers for BD outreach sequencing:

- Tier 1 (score ≥ 80) — high priority, targeted for in-person outreach in Week 1
- Tier 2 (score 50–79) — medium priority, targeted for phone outreach in Weeks 2–3
- Tier 3 (score < 50) — lower priority, suitable for email and digital campaigns in Month 2+

7. Canvassing Strategy — Planned Approach

7.1 Territory Design

Tier 1 merchants will be grouped into suburb clusters for BD rep assignment. Initial zones planned based on KMF geo scores:

- Zone A: Hurstville, Kogarah, Rockdale — high F&B and Beauty density, strong KPay cultural fit
- Zone B: Parramatta, Merrylands, Auburn — mixed Retail and F&B, high population density
- Zone C: CBD, Surry Hills, Alexandria — high transaction volume, lower cultural fit but high merchant density

7.2 Outreach Cadence

- Week 1: Tier 1 in-person visits, prioritised by suburb cluster
- Weeks 2–3: Tier 2 phone outreach
- Month 2: Tier 3 email and digital campaigns
- Referral loop: signed Tier 1 merchants are asked to refer neighbouring businesses, compounding BD efficiency

7.3 Pitch Angles by Segment

- F&B in Asian-Australian suburbs: 'Your customers already use WeChat Pay and UnionPay — accept them in one terminal'
- Beauty and nail salons: 'Same-day settlement means you see your money tonight, not in 3 days'
- Independent retail: '\$0 terminal, no monthly fee — switch in 10 minutes, save ~\$1,200/year'

8. Performance Dashboard — Planned Metrics

A dashboard will be designed for the sales manager to track the canvassing initiative. The five planned metrics are selected to track revenue pipeline progress, not just BD activity:

- Estimated Monthly GMV in Pipeline (\$) — hero metric, top of dashboard. Tier 1 merchants contacted × average GMV by sector. This is the number the Growth Manager opens every Monday.
- Tier 1 Coverage % — what percentage of Tier 1 merchants have been contacted this week. Creates urgency and tracks BD sprint progress.
- Contact → Meeting Rate — flags whether the phone quality issue is still affecting BD outreach. If this drops, it signals more Level 1 phones in the current outreach batch.
- Meeting → Sign Rate by Sector — reveals which sector_level_2 categories convert best, allowing BD to prioritise sub-sectors within Tier 1.
- Average Days to Sign — sales cycle length for pipeline forecasting and resource planning.

9. Tools & Environment

Language	Python 3.11 (conda environment: kpay_case_study)
Core libs	pandas, phonenumbers, rapidfuzz, matplotlib, seaborn
Notebook	Jupyter Notebook — Case-Study-Clearning-Enrichment.ipynb
Dashboard	Tableau
Presentation	PowerPoint
Version ctrl	Git repository — Kpay_caseStudy
Hardware	MacBook Pro M2 Pro, 16GB RAM

10. Planned Timeline

Given the March 1 deadline, the work is scoped into must-have and nice-to-have components:

Must-Have (committed scope)

- Data audit + quality risk table with BD impact framing
- Phone classification (7 types), quality levels, and reachability scoring
- Country detection and AU/non-AU separation
- Suburb extraction from address column
- Sector remapping and Others rescue
- KMF scoring + tier assignment
- BD canvassing strategy
- Dashboard blueprint with 5 KPIs and rationale
- 10-slide presentation deck

Nice-to-Have (if time allows)

- Suburb heatmap visualisation (folium)
- QA appendix — 6-point data validation table
- 'If I Had More Time' slide: ML lead scoring, deeper data analysis, dashboard enrichment